

DoS Stress-Testing & Auto-Mitigation for AI Inference Servers

Cost-aware admission control in front of a vLLM endpoint

`Team name` · `Member 1` `Member 2` `Member 3`

Target: vLLM 0.29.0 · Qwen2.5-0.5B-Instruct · NVIDIA Tesla T4

`github.com/mena-04/DoS-Stress-Testing`

MEASURED ON THE GPU, NOT SIMULATED

Under an eight-way flood of 512-token generations, turning the gateway on took legitimate p95 latency from **9.55 s to 1.32 s** while *increasing* the number of legitimate requests answered from **3 to 15** in the same 20-second window. **No legitimate request was rejected in either arm.** Latency and throughput improved together, which is what distinguishes protecting users from simply refusing them.

7.2×

lower legitimate p95

5.0×

more legitimate requests
served

0 → 50%

attacker rejection rate

0%

legitimate traffic rejected

1. Project objective

The unit of load on an inference server is not the request — it is the token. Every rate limiter in common use disagrees. `nginx limit_req`, API-gateway requests-per-second rules and WAF throttles all count requests, because for a conventional web service requests cost roughly the same. On an LLM endpoint they do not: a 24-token chat reply and a 2048-token generation differ in GPU cost by nearly three orders of magnitude, and both are a single well-formed request.

That gap is the whole attack. On our T4 we measured decoding at **0.035 s per token**, so one 1024-token generation occupies a vLLM sequence slot for roughly **36 seconds**. The server was launched with `--max-num-seqs 8`. Eight requests — a rate that no sane requests-per-second limit would block, and that no anomaly rule would call a flood — therefore hold the entire GPU for half a minute. The attacker needs no botnet, no spoofing and no volume. It needs a large `max_tokens`.

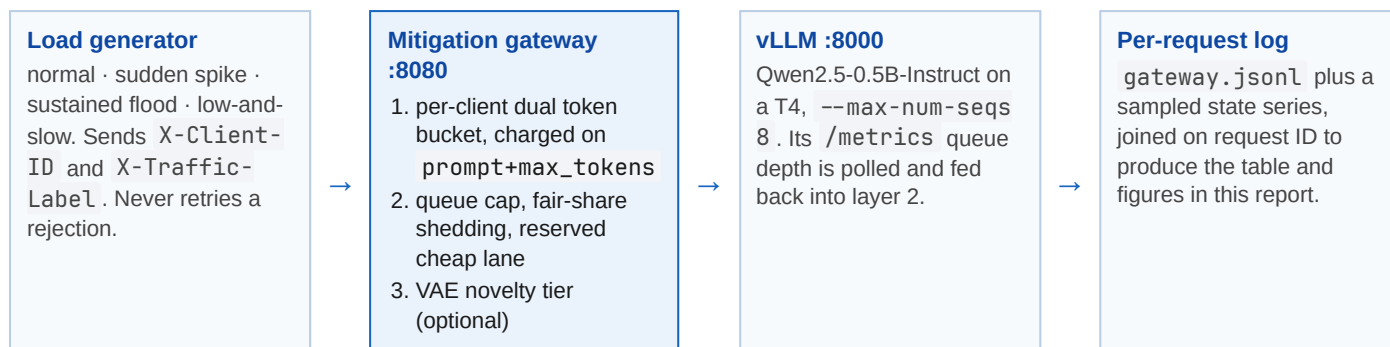
Who deploys this, and what it replaces

The user is a team self-hosting an open-weights model (vLLM, TGI, Ollama) behind a public or partner-facing API, without per-tenant GPU isolation — which is the normal situation, because dedicating a GPU per tenant is what self-hosting is meant to avoid. Our gateway occupies the position an API gateway already occupies, speaks the same OpenAI-compatible protocol, and replaces the request-per-second rule living there with a *token-cost budget*. Nothing about the model, the server or the client changes: the endpoint URL moves by one port.

Two things follow from that placement, and both shaped the design. Rejection has to be *immediate and legible* — a caller that waits 30 seconds for a 503 has already been denied service, so every refusal carries an `X-Gateway-Reason` header and a `Retry-After`. And the mitigation must be *measurable*: the reason a team tolerates a component in front of its inference server is evidence that it helps, which means the gateway's job includes producing the before/after numbers in this report.

2. Proposed solution

A FastAPI reverse proxy in front of vLLM, carrying three admission layers. Every decision is written to `gateway.jsonl` with its reason, so the analysis reads the gateway's own record of what it did rather than inferring it from latency.



Mechanism 1 — per-client dual token bucket (cost, not rate)

Each client gets two buckets. The first meters request rate, as any limiter would. The second meters **cost**, and is charged `prompt_tokens + max_tokens` **at admission, before generation starts**. Charging up front is the load-bearing detail: the true cost is only known when a generation finishes, and by then the GPU time is spent, so the limiter must bill the declared intent. Three corollaries follow, each a way an attacker would otherwise get in free: an omitted `max_tokens` is charged the server's default, not zero, since the model will generate to the context limit; `n` and `best_of` multiply the charge because they multiply the decode work; and a request costing more than the whole bucket is refused as `cost_exceeds_budget` rather than told to retry, since no wait would ever admit it.

Mechanism 2 — queue cap with fast rejection

`global_max_inflight` is held **at** the backend's `--max-num-seqs`. This is the single most important number in the system and the least obvious. Admit more than vLLM can batch and the queue still forms — but it forms *inside* vLLM, where the gateway can no longer see it, reorder it, or shed from it. We measured exactly that: with 48 admission slots against a backend accepting 8, gateway pressure read zero for an entire run while legitimate p95 reached 4.0 s. Beyond the cap, a request waits a bounded 200 ms for a slot and is then refused with a 503. Pressure is evaluated from four signals — local occupancy, a latency EWMA, and vLLM's own `num_requests_waiting` and `num_requests_running` — each with separate high and low water marks, so the controller cannot flap between shedding and admitting on every tick. Under pressure, shedding is *fair-share*: clients are ranked by their share of recent token cost and only those above an equal share are refused, which is what keeps a quiet client flowing while a flooder absorbs the rejections.

Mechanism 3 (optional layer) — VAE novelty detection

A variational autoencoder over a 12-feature per-client behavioural window (request and cost rate, prompt-size distribution, inter-arrival mean and spread, concurrency, prompt repeat ratio, error rate, latency). It is trained **on normal traffic only** and scores reconstruction error as novelty, so no attack is ever labelled for it and there is no leakage between the attack profiles and the detector that flags them. Score tiers tighten a client's bucket multipliers rather than blocking outright, decay instead of latching, and fail open.

Three rules that protect the measurement, not the server. (1) The gateway never reads `X-Traffic-Label`. The label is logged for the analysis join and is invisible to admission control — branching on ground truth would turn "attacker rejection rate" into a measurement of the label rather than of the defence. **(2)** Mitigation-off still goes through the gateway. `configs/off.yaml` is a pure passthrough that still logs, so the proxy hop appears in both arms; a baseline that bypassed the gateway would credit the mitigation with removing a cost it never added. **(3)** The load generator never retries a 429 or 503. A retrying client converts rejection into amplification and makes the rejection rate meaningless.

Why this approach now

Two things make cost-aware admission practical today that were not available a few years ago. vLLM's continuous batching and paged KV cache mean the scarce resource is *sequence slots and KV blocks* — a countable quantity with a known limit — rather than diffuse CPU time. And vLLM exports `num_requests_running` and `num_requests_waiting` on a Prometheus endpoint, so a proxy can read the backend's real queue instead of guessing at it from response times. A design predating that telemetry had to infer saturation from latency alone, which is a lagging signal: by the time p95 moves, the queue is already deep. We use latency as one of four inputs, and the other three are structural.

The same shift is why a purely learned detector would be the wrong primary mechanism here. Load shedding has to be correct on the first request of an attack it has never seen, and a model cannot promise that. So the two required mechanisms are deterministic and auditable, the VAE only adjusts their parameters, and the whole system degrades to the deterministic pair if the model is unavailable.

3. Solution validation

Component	What was actually run
Target service	vLLM 0.29.0, Qwen2.5-0.5B-Instruct, <code>--dtype half --max-model-len 2048 --gpu-memory-utilization 0.85 --max-num-seqs 8</code> , Tesla T4 16 GB. Real GPU, real weights, real Prometheus metrics.
Load profiles	Normal, sudden spike, sustained flood and low-and-slow (Locust), plus a two-client probe that drives a cheap legitimate stream against an expensive attacker stream.
Mitigation arms	Four configs selected by one flag: <code>off</code> , <code>ratelimit</code> , <code>ratelimit_queue</code> , <code>full</code> . The ablation isolates which mechanism produced which effect.
Automated tests	75 unit and integration tests, including end-to-end runs over real sockets against a backend stand-in.
GPU-free reproduction	<code>gateway.fake_vllm</code> reproduces the measured T4 timing (0.035 s/token decode, 0.0026 s batch penalty, 0.0002 s/token prefill) and vLLM's metric shape, so the whole experiment — including saturation — runs on a laptop and in CI.

Four negative results, and what each one changed

The mitigation was the easy half. Most of our effort went into tests that appeared to pass while measuring nothing, and each one forced a design change.

(a) The first overload test did not overload anything. Fifty concurrent "expensive" requests returned p50 5.603 s, p95 5.700 s, max 5.701 s. Those three values lying within 0.1 s of each other is the tell: every request finished in the same batch, so no queue ever formed and the run was a measurement of batch latency, not of saturation. Two causes. Automatic prefix caching is on by default in vLLM's V1 engine, and all fifty prompts were identical, so forty-nine of them skipped prefill almost entirely. And KV capacity was never the constraint: at 24 layers $\times$ 2 KV heads $\times$ 64 head-dim $\times$ 2 $\times$ 2 bytes, this model spends about 12 KB per token, so ~11 GB of KV cache holds ~900k tokens against a demand of ~100k — a 9 $\times$ margin. *Changed:* randomised prompts, prefix caching disabled for load runs, and a concurrency sweep to locate the knee rather than assuming one exists at 50.

(b) The queue-depth series was all zeros, and the server looked healthy. The metrics sampler scraped `/metrics` inside a bare `except: pass`, and read a missing value as `0`. Under load, `/metrics` is exactly what stops responding — so the moment the queue got deep, the recorded queue depth went to zero. *Changed:* the gateway's pressure controller treats an absent upstream snapshot as *unknown* and never as empty; it falls back to the two signals it owns locally, and staleness is recorded per sample and per request so a gap in the data is visible as a finding rather than smoothing into a flat line.

(c) Identity rotation defeated every per-client defence. Splitting the same flood across eight client IDs produced zero rejections and a legitimate p95 of 4.0 s. This is not a bug to patch; per-client buckets dilute under rotation by construction, and any limiter keyed on identity has the same hole. *Changed:* a reserved admission lane partitioned on **request cost** rather than identity. Cheap requests can always use it, expensive ones never can, and an attacker gains nothing from rotating IDs because the partition does not ask who is calling.

(d) Fair-share inverted under long-running requests. Cost is charged at admission and aged out of a 10-second window, but a 512–1024-token generation runs far longer than that. So an attacker holding all eight slots aged out to a recorded cost of `0.0`, while the steady cheap client — the only one with *recent* cost — measured as 100% of the window and was shed: legitimate traffic took **45.5% rejection** at `cost_share=1.0`, with the flooder refused for nothing. Precisely backwards, and worst on the low-and-slow profile, which is built out of long-lived requests on purpose. *Changed:* in-flight cost stays charged until the request completes, is released on completion rather than held as a penalty, and is dropped entirely when a client has nothing running, so a lost completion cannot charge a client forever.

Statistical care

- **Latency is never reported without throughput beside it.** A gateway that rejects 95% of traffic has an excellent p95, because shedding removes load from the latency sample. Our headline p95 fell 7.2 $\times$ *while* legitimate throughput rose 5.0 $\times$ and legitimate rejection stayed at zero — three facts that cannot all hold if the gain came from refusing users.
- Percentiles are nearest-rank over *successful* requests only; a fast 503 is a failure, not a quick success, and is counted in the rate columns instead.
- The VAE sees only normal traffic in training and scores novelty, so no attack profile appears on both sides of the train/test boundary.
- One invalid run was **excluded and recorded** rather than quietly dropped: a pass in which every request returned `ConnectError` because no gateway was listening. Its 35,128 attacker "requests" are refused connections returning instantly, not load.

4. Results

Both arms below are the same load against the same live vLLM server on the same T4, differing only in the config the gateway was started with: eight concurrent 512-token attacker requests from a single identity, against a legitimate client issuing one 32-token request every 0.4 s, for 20 seconds.

Metric	Mitigation off	Mitigation on	Change
Legitimate p95 latency	9.546 s	1.323 s	7.2× lower
Legitimate p50 latency	8.867 s	0.894 s	9.9× lower
Legitimate requests answered	3	15	5.0× more
Legitimate success rate	100%	100%	—
Legitimate requests rejected	0	0	—
Attacker requests admitted	24	8	3.0× fewer
Attacker rejection rate	0%	50%	+50 pts
Attacker p95 latency	10.355 s	12.113 s	1.17× higher
Rejection reason	—	client_concurrency ×8 (429)	

Legitimate users during an 8-way flood (0 legitimate requests rejected in either arm)

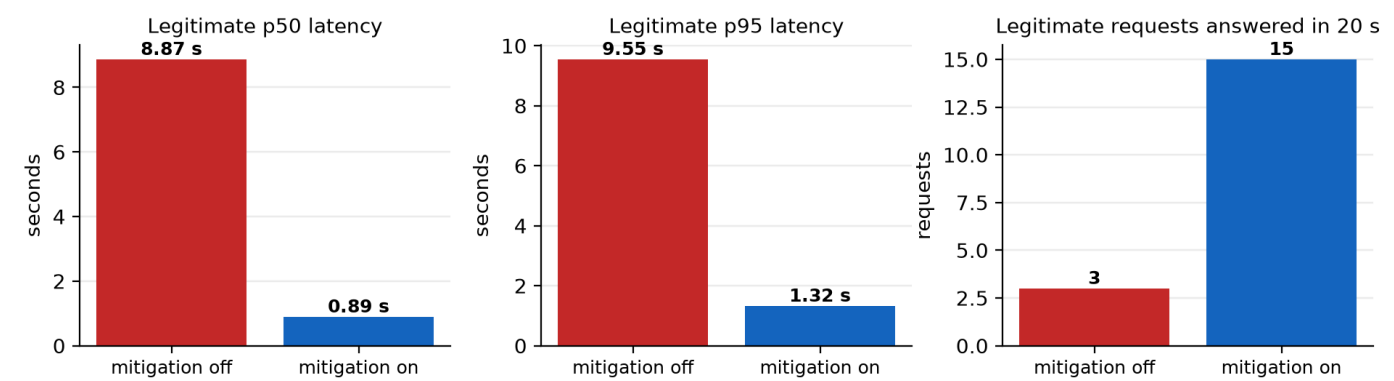


Figure 1. The two panels on the left are the result; the panel on the right is what makes it credible. Latency fell and the number of requests served rose at the same time, with zero legitimate rejections in either arm, so the improvement is not shedding in disguise.

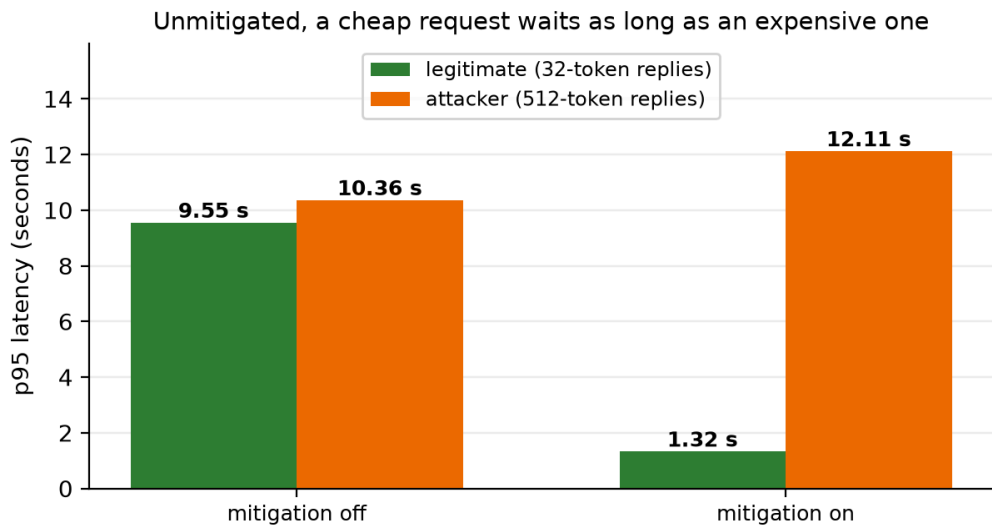


Figure 2. The failure mode, stated directly. Unmitigated, the cheap request and the expensive one complete in almost the same time (9.55 s vs 10.36 s) — one shared queue charges a 32-token reply the same wait as a 512-token generation. With cost-aware admission the two separate by 9×: the caller asking for more work waits for it.

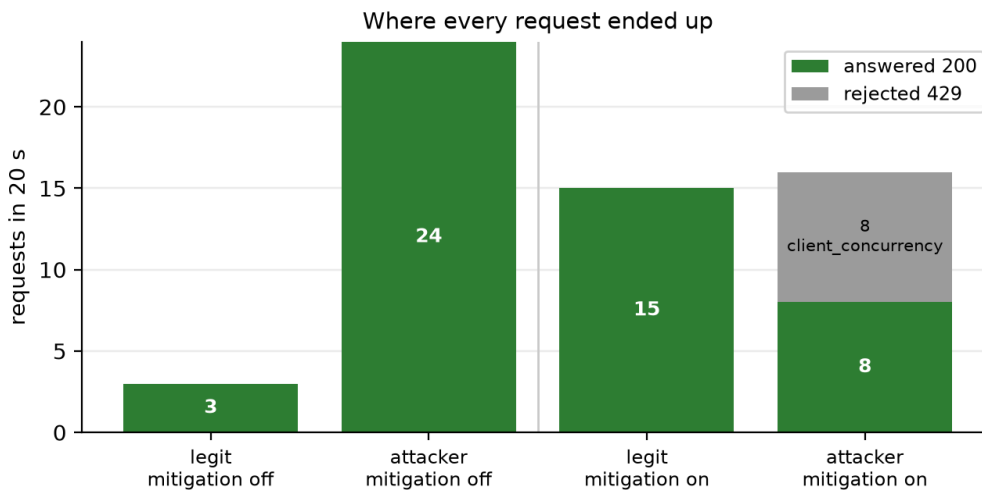


Figure 3. Every request in both arms, as counts rather than rates. Unmitigated, the attacker gets 24 full generations and the legitimate client gets 3 slow ones. Mitigated, the attacker is cut to 8 and refused 8 with an explicit reason, and that reclaimed capacity is what the legitimate client's 15 answered requests are made of.

Interpretation

The capacity did not appear from nowhere and the total work done barely changed — it was *reallocated*. Unmitigated, eight expensive generations hold all eight sequence slots, and a 32-token request arriving behind them waits for one to finish; that is the 8.9 s. The gateway's cap keeps the queue on its own side of the wire, where the reserved cheap lane and fair-share ranking can order it, so a small request is no longer stuck behind a large one. Note that the attacker's own p95 rose, from 10.36 s to 12.11 s, while half its requests were refused outright. Being an expensive client became slower and less reliable, which is exactly the incentive a cost-based limiter is supposed to create.

Limitations we can state precisely

- **This run used a single attacker identity.** The rejections are `client_concurrency` — a per-client cap, which is the one mechanism that identity rotation defeats (finding c). The cost-partitioned reserved lane exists for the rotating case and is verified against the calibrated stand-in, *not yet on the T4*. The honest headline is therefore: proven on the GPU against a single-identity flood, proven in simulation against a rotating one.
- **Small samples.** Three and fifteen legitimate requests in 20 seconds. The p50/p95 separation is many times larger than the observed spread (max equals p95 in three of the four series), so the direction is not in doubt, but a longer run is needed before quoting a confidence interval.
- **The four Locust profiles were run against vLLM directly**, before the gateway was in the path. They characterise the backend and calibrate the stand-in; they do not yet measure mitigation effect per profile. Re-pointing them at port 8080 is the immediate next step and needs no new code.
- **Thresholds are matched to this backend, not tuned.** `global_max_inflight` tracks `--max-num-seqs`, but the latency water marks still assume a cheaper request mix than a 1024-token ceiling produces, which leaves pressure reading ELEVATED for most of a flood. Calibration follows the concurrency sweep.

Conclusions

Rate limiting the wrong quantity provides no protection at all. Eight requests, within any plausible requests-per-second budget, took a GPU-backed endpoint from sub-second to nine-second responses. Charging `prompt_tokens + max_tokens` against a per-client budget at admission time, and capping concurrent admissions at the backend's real batch width, restored a 1.3 s p95 and served five times as many legitimate requests without refusing a single one.

Hold the queue where you can still act on it. Admitting more than `--max-num-seqs` does not remove the queue, it only moves it somewhere the mitigation cannot reach. This one alignment mattered more than any threshold we tuned.

Partition on cost, not on identity. Every identity-keyed defence dilutes under rotation, and ours did — eight IDs, zero rejections. The mechanism that survived was the one that never asks who is calling.

The measurement is as easy to get wrong as the mitigation. Our first flood did not overload the server, and our first queue-depth series read zero precisely when the queue was deepest. Three of the four design changes above came from a test that proved nothing while appearing to pass — which is why the gateway records its own decisions, its own reasons, and the staleness of the signals it acted on.